ASSIGNMENT

UCSH- 301

COMPUTER ORGANISATION & ARCHITECTURE

Topics Covered:-

Gates, Combinational Circuits and Sequential Circuits

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:- Group-2 Assignment :-

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GATES

A Gate is an electronic circuit that produces an output signal that is simple Boolean operation on it's input signals.

In Layman terms, it means "single output by multiple inputs".

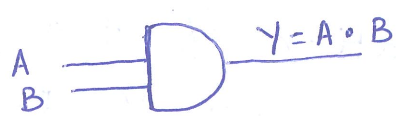
The basic gates used in digital logic are: AND, OR, NOT, NAND, NOR and XOR.

AND gate:- (•)

It is denoted by "•" symbol. The function is given by $Y = A \cdot B$.

The truth table for AND gate is as follows:

A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1



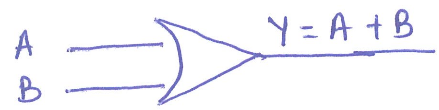
Graphical Representation

OR GATE :- (+)

It is denoted by "+" symbol. The function of OR gate is $Y = A + B$.

The truth table is as follows:-

A	B	$Y = A+B$
0	0	0
0	1	1
1	0	1
1	1	1

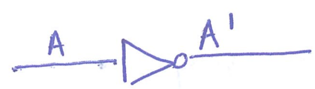


Graphical Representation

NOT GATE :-

It is also known as compliment or inverter. The truth Table is as follows:-

A	A'
0	1
1	0



Graphical Representation

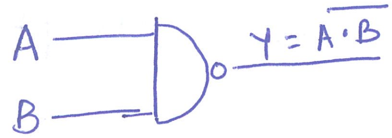
NAND GATE :-

Combination of two gates AND, NOT.
The function of this gate is as follows: $Y = \overline{A \cdot B}$.

The truth of NAND gate is given below:-

P.T.O.

A	B	$Y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0



Graphical Representation

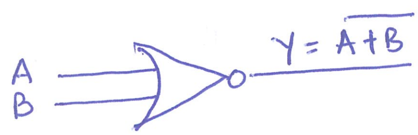
NOR GATE :-

Combination of two gates OR, NOT. The function of this gate is as follows:

$Y = \overline{A + B}$.

The truth of NOR gate is as follows:

A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0



Graphical Representation

⇒ The NAND and NOR gates are also called universal gates and the three basic gates :- OR, AND, NOT can be derived from these two gates.

P.T.O.

Some of the other gates which have use in this chapter are:- XOR and XNOR.

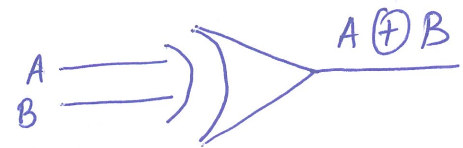
XOR:- (Exclusive OR).

This gate is also known as "odd no of one's operator". It is called so because, wherever inputs have odd numbers of one's, the output is 1 else it is 0.

The function is given by:-

$$Y = A'B + AB' = A \oplus B.$$

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

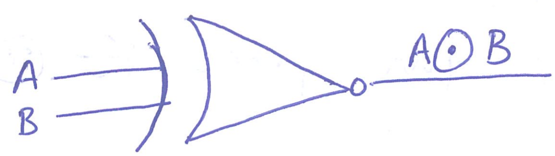


Graphical representation

XNOR:- (Exclusive NOR):

Similar to previous gate, this gate also same but with NOR. So, the function is given by $Y = A'B' + AB = A \odot B.$

A	B	$A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1



Graphical Representation

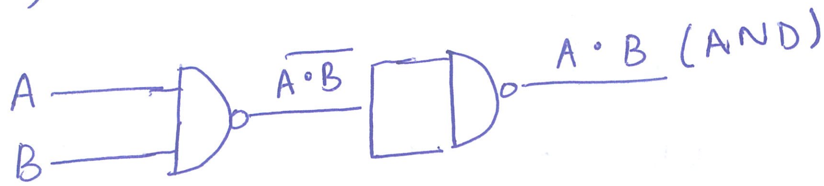
Q) Prove NAND and NOR gates are universal gates.

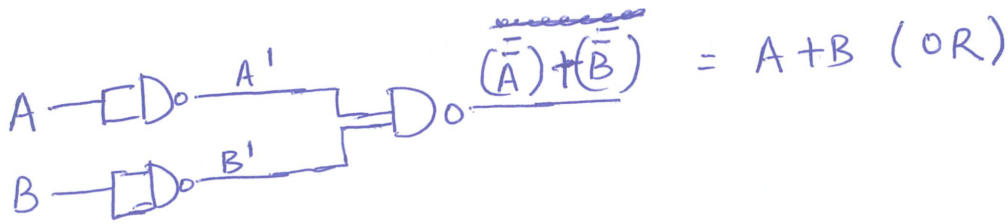
ans:- <u>NAND gate:-</u>

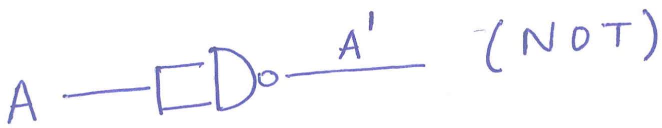
(i) $(\overline{A \cdot B})' = A \cdot B$ (AND)

(ii) $\overline{\bar{A} \cdot \bar{B}} = (\bar{A})' + (\bar{B})' = A + B$ (OR)

(iii) $\bar{\bar{A}} = A$ (NOT).





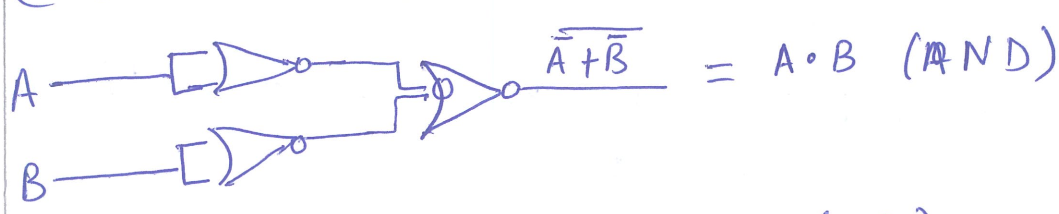


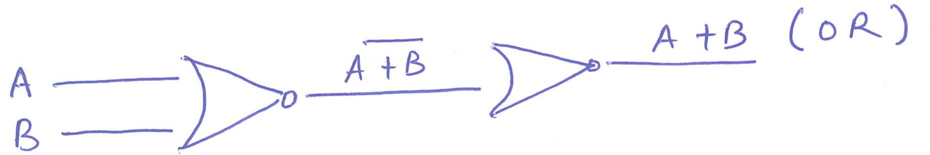
<u>NOR gate:-</u>

(i) $(\overline{A + B})' = A + B$ (OR)

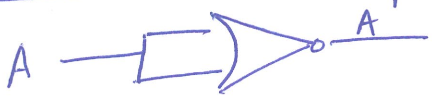
(ii) $(\bar{A} + \bar{B})' = (A')' \cdot (B')' = A \cdot B$ (AND)

(iii) $\bar{\bar{A}} = A$ (NOT)





NOT proof:-



(NOT).

COMBINATIONAL CIRCUITS

Combinational circuits are those circuits in which the output is dependent only on present input.

Some examples are :- Half adder, full adder, subtractor, full subtractor, parallel adder subtractor, multiplexer, decoder, encoder etc.

Procedure to design any Combinational Circuit:-

(i) What is desired circuit?
(ii) Principle behind the circuit.
(iii) No of inputs and outputs.
(iv) Block diagram of circuit.
(v) Truth table for all the possible combinations in circuit.
(vi) Draw K-map from truth table.
(vii) obtain minimised Boolean function of desired circuit.
(viii) Draw the circuit diagram.

Note:- In the following circuits, which will be discussed, the above procedure will be used.

HALF ADDER

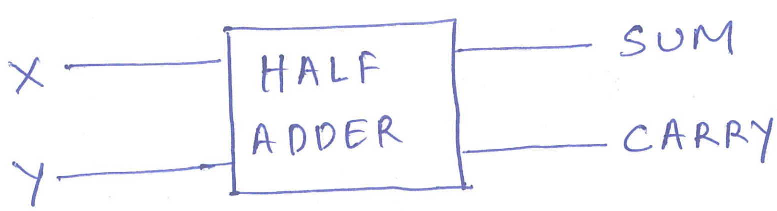
A half-adder is a circuit that is used to add two single bits.

Principle:- It uses XOR gate and AND gate to add two binary numbers and procedure two outputs i.e, sum, CARRY.

X	Y	F=X+Y	CARRY
0	0	0	
0	1	1	
1	0	1	
1	1	0	1

No of inputs: 2
No of outputs: 2.

Block-diagram:-



[Black Box Rotation]

Truth Table:-

X	Y	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

K-Map:-

for sum

A \ B	0	1
0	0	1
1	1	0

$SUM = AB' + A'B$

$= A \oplus B$

for Carry

A \ B	0	1
0	0	0
1	0	1

$CARRY = AB$

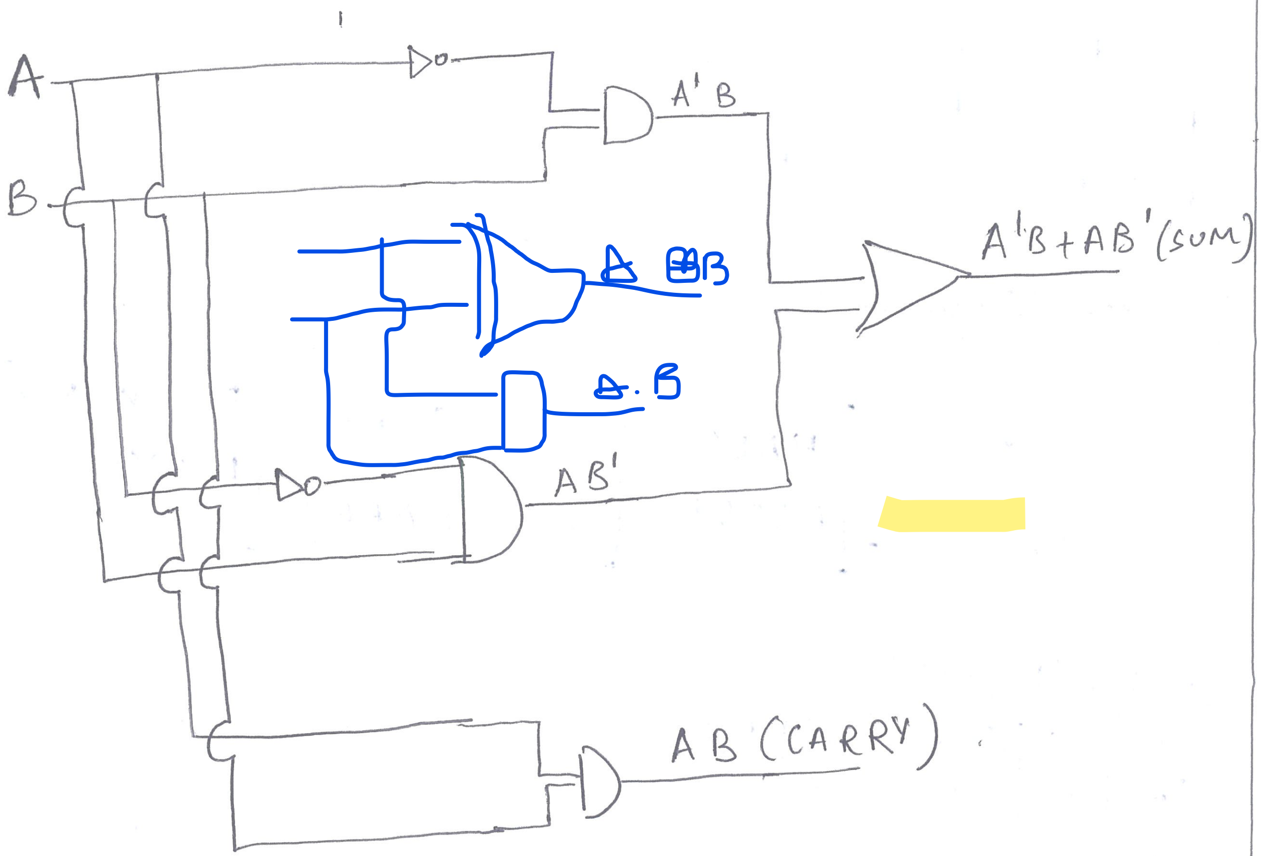
$= A \cdot B$.

Reduced Boolean expression:-

$SUM = A'B + AB'$

$CARRY = AB$.

Circuit Diagram:-



(HALF ADDER).

FULL ADDER

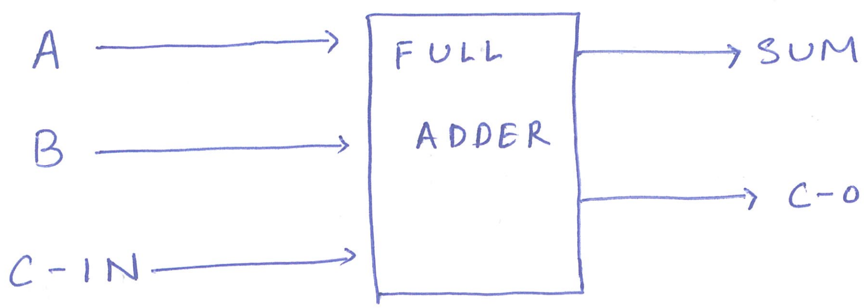
A combinational circuit that can add two binary digits and a carry bit, and produces a sum bit and carry bit as output is called Full-adder.

Principle:-

Add three binary bits and produce two outputs i.e, SUM and CARRY.

No of inputs:- 3
No of output:- 2

Block Diagram:-



Truth-Table

A	B	C-IN	SUM	C-OUT
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

two or more ones carry-out is one

K-MAP:-

SUM

A \ BC	00	01	φ1	10
0	0	1	0	1
1	1	0	1	0

$SUM = A \oplus B \oplus C_{in}$

$CARRY_{out}$

A \ BC	00	01	11	10
0	0	0	1	0
1	0	1	1	1

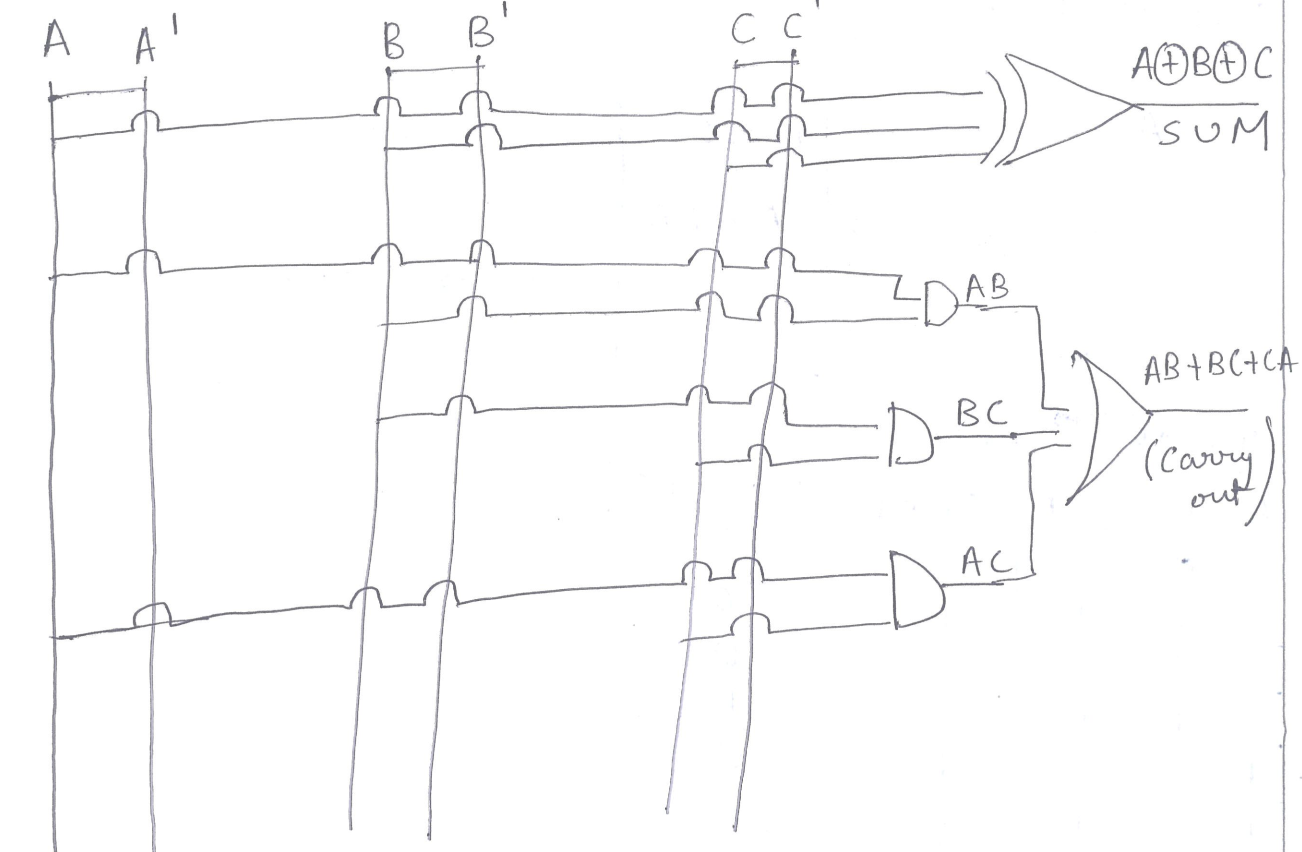
$CARRY_{out} = AB + BC_{in} + C_{in}A.$

(~~redundancy~~)

Reduced Boolean expression:- $A \oplus B \oplus C_{in}$ (SUM)

$AB + BC_{in} + C_{in}A$ (C-OUT).

Circuit diagram:-

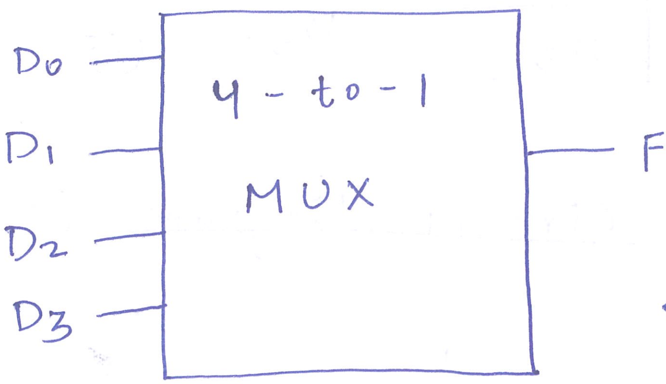


(FULL ADDER).

Multiplexers:-

The multiplexer connects multiple inputs to single output.

At any time, one of the inputs is selected to be passed as the output.



Block diagram

This represents a 4-to-1 multiplexer. There are 4-input lines D_0, D_1, D_2 and D_3. One of these lines is selected to provide the output signal F.

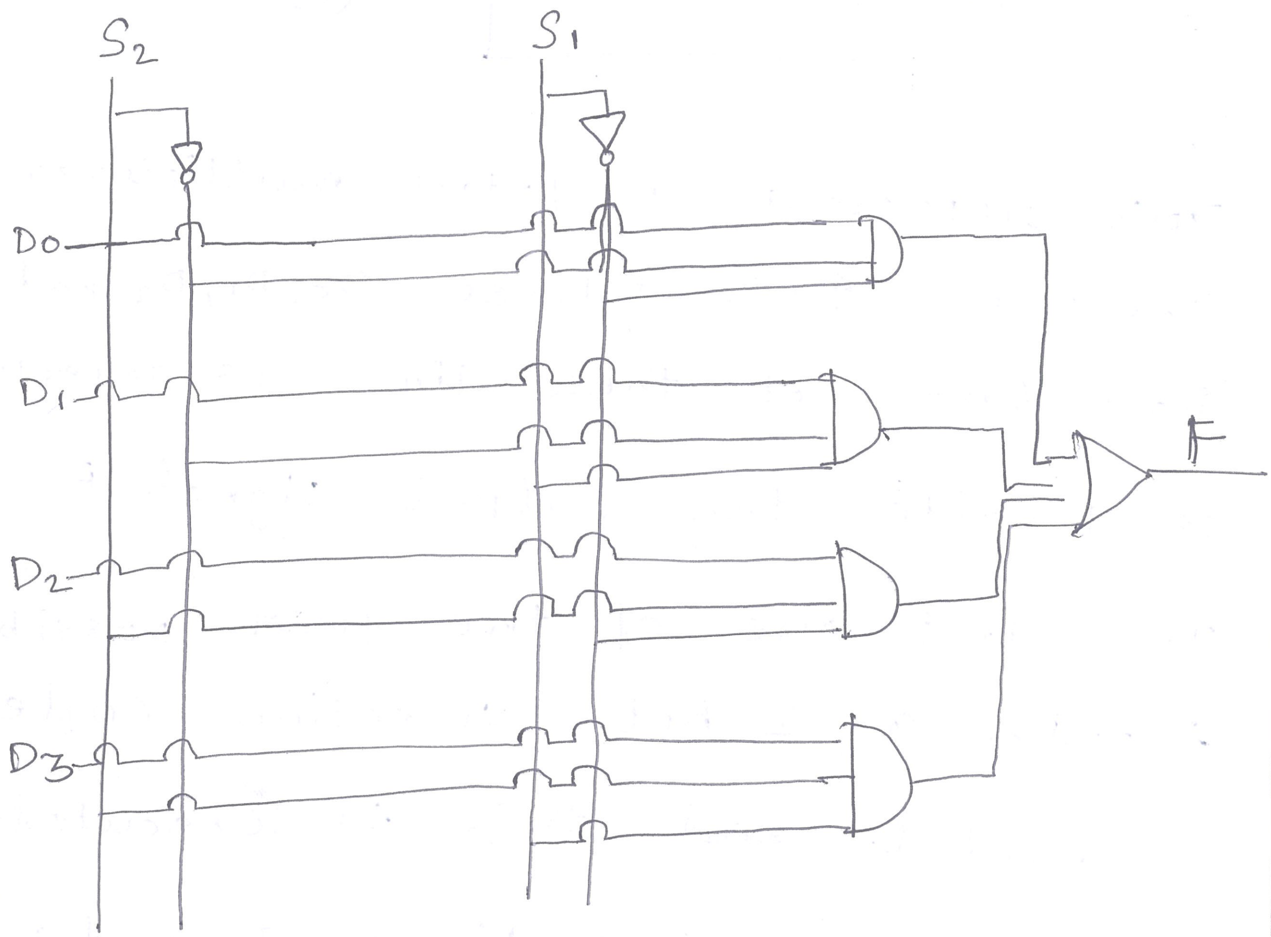
To select one of the four possible inputs, a 2-bit selection code is needed, and this is executed using two select lines S_1 and S_2.

Truth Table:-

S_2	S_1	F
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

This is simplified truth table. Instead of showing all possible combinations, it shows output as data from line D_0, D_1, D_2 D_3. ☺

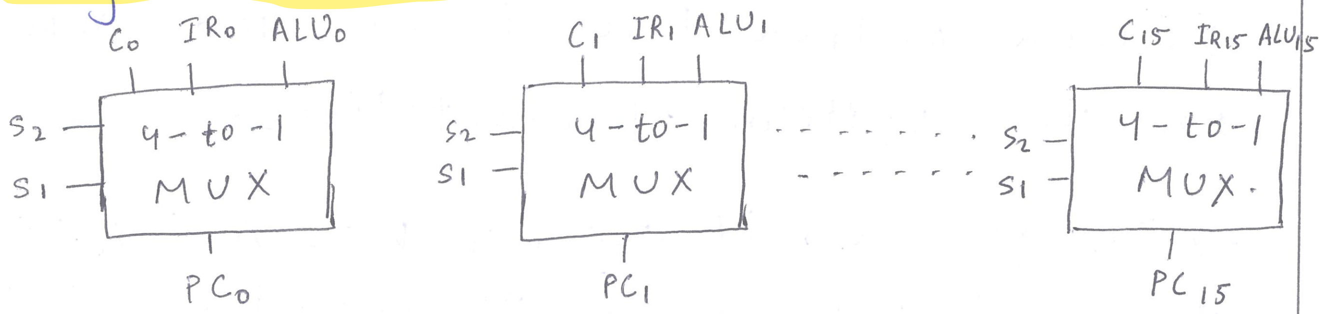
Multiplexer Circuit diagram:- (4 to 1 MUX).



(4 to 1 MUX).

Multiplexer Application:-

Multiplexers have wide range of applications one of them is "Multiplexer Input to Program Counter".



The above diagram is the basic representation for "Multiplexer input to Program Counter".

→ A binary Counter, if PC is to be incremented for next instruction.

→ The instruction register (IR), if a branch instruction using a direct address has just been executed.

→ The output of ALU, if the branch instruction specifies the address using displacement mode.

Because PC contains multiple bits, multiple multiplexers are used. "Interesting point" ☺

Decoders:

A Decoder is a Combinational circuit with a number of output lines, only one of which is asserted at any time.

→ which output line is asserted depends on the pattern of input lines.

→ In general, if decoder has n inputs then it will have 2^n outputs.

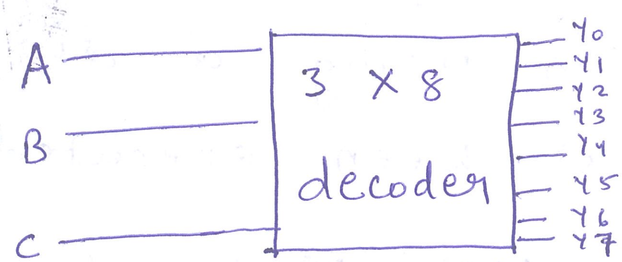
Relationship :-

Let m be no of inputs, n be no of outputs

then, $n = 2^m$.

Now, we desire to make 3 input decoder so, 8 outputs.

Block diagram:-

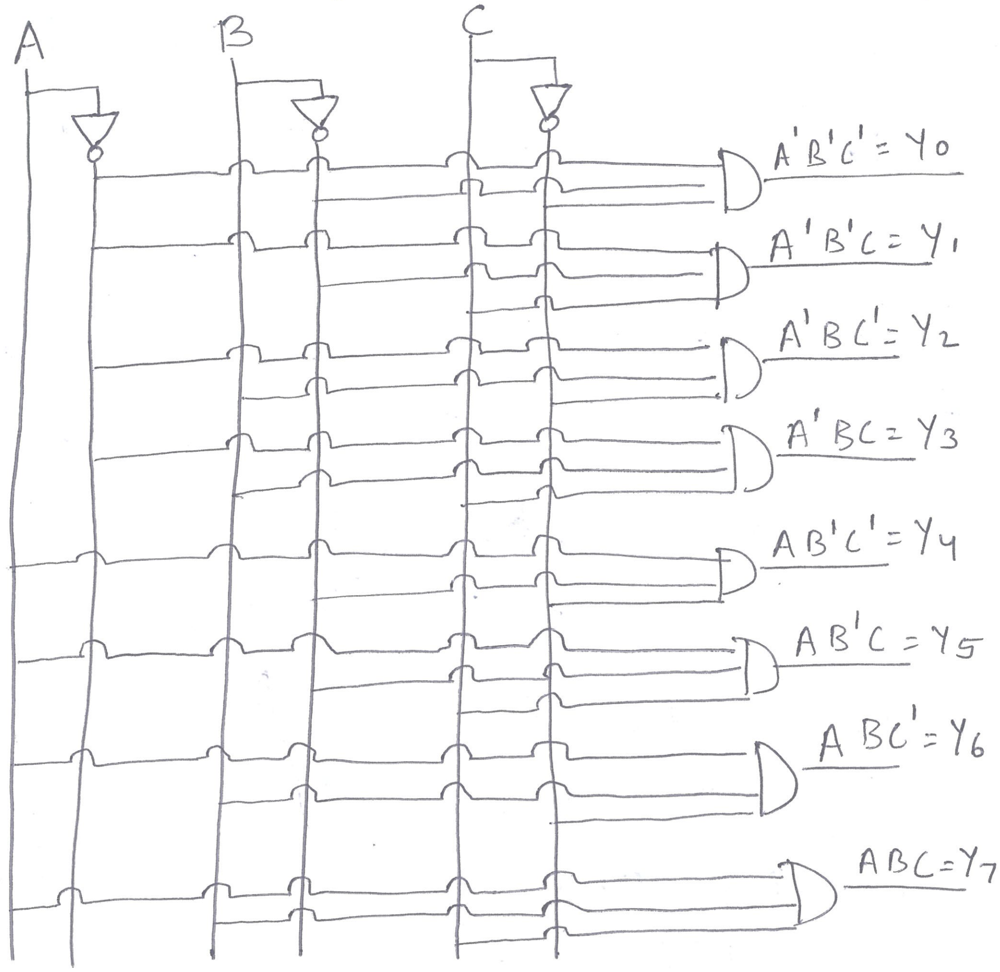


If locations are 2^m, address bytes is n.

Truth Table :-

A	B	C	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

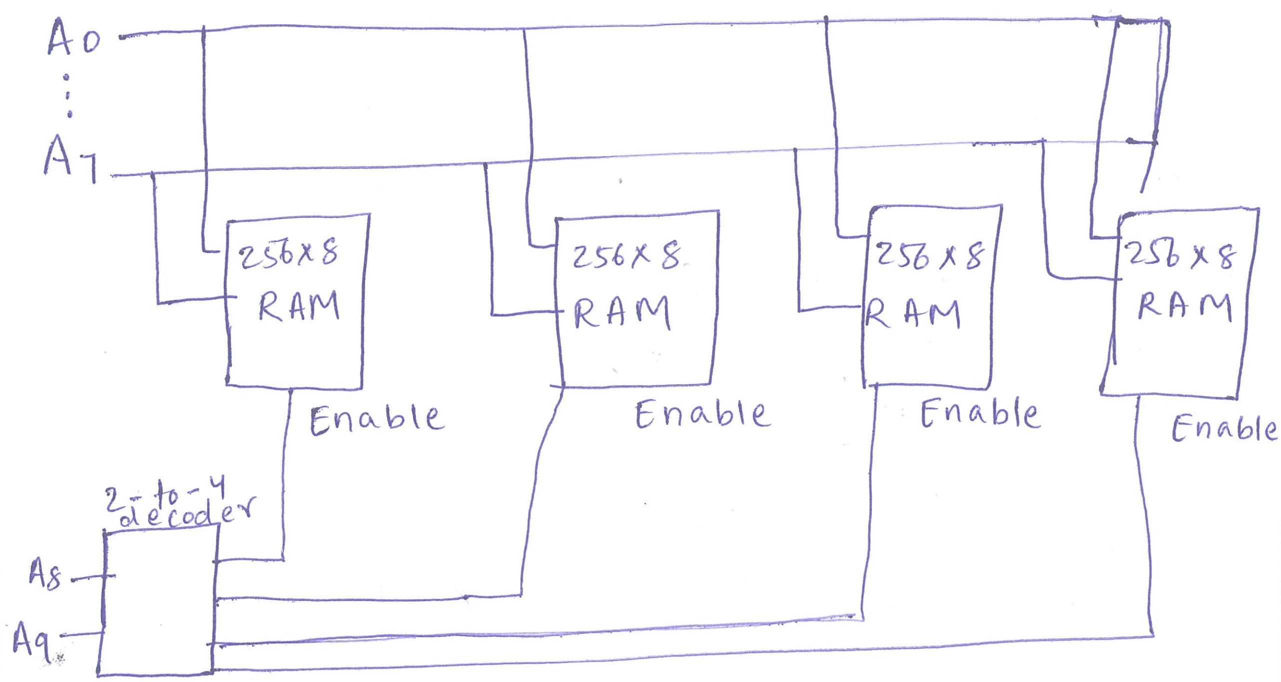
Circuit diagram :- (3 x 8 Decoder)



Application of Decoder:-

→ Decoders are used to fetch the location of digital signal system.

• **256 X 8 RAM:-** It is a memory chip consisting of 256 memory locations with each location consisting of 8 bytes.



In this diagram, the no of address lines are 10 ($A_0 - A_9$).

→ In this $A_0 - A_7$ are used for addressing memory chip locations.

→ A_8 and A_9 are used as input to a 2 x 4 decoder.

→ Higher order address bits are always used as input into the decoder circuit for selecting a memory chip (CS) (Chip Select).

SEQUENTIAL CIRCUITS

→ A sequential circuit is a circuit in which output is not only dependent on present input but also on previous outputs.

Ex :- Timer, Counter.

Flip-Flops :-

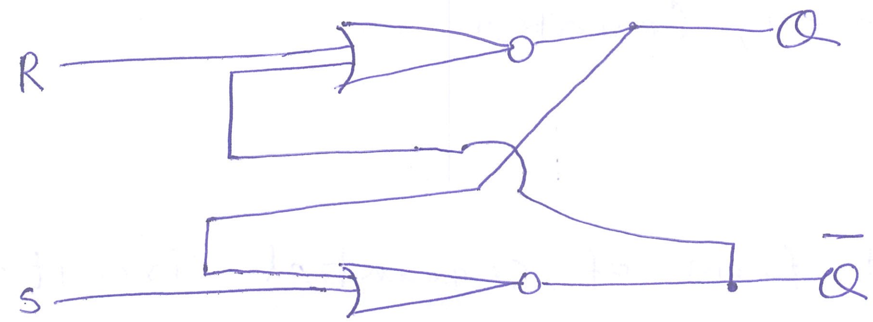
→ Simplest form of sequential circuit.

→ There are two important properties of sequential circuit.

(i) The flip-flop is bistable device. It exists in one of two states and in absence of input, remains in state. Thus, the flip-flop can function as 1-bit memory.

The flip-flop has two outputs, which are always the complements of each other.

<u>S-R LATCH</u>:- It's also called S-R flip-flop.. The 'S' is for "set" and "R" is for "Reset". and two outputs Q and $\bar{Q}$ and Consists of two NOR gates connected in feedback arrangement.



<u>NOR-LATCH</u>:-

CONCLUSION:- If one of inputs is 1, output is 0. this can be understood by

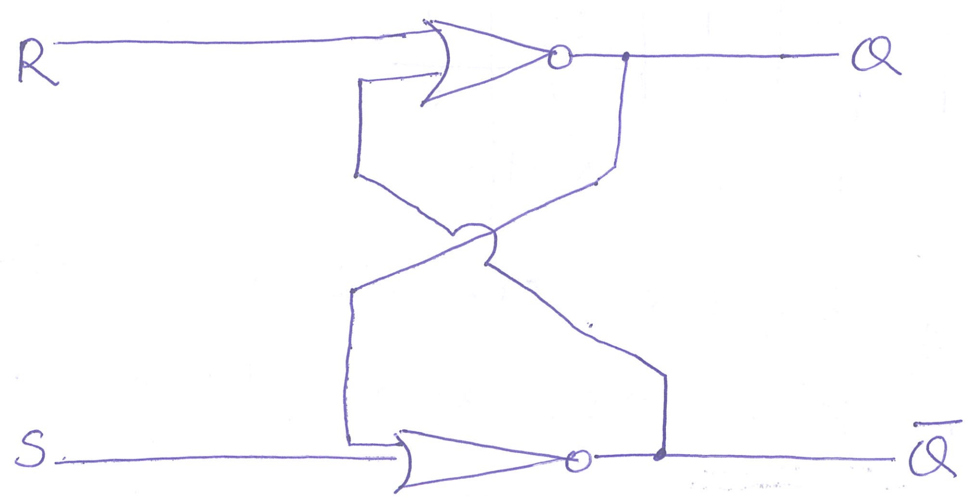
CASE 1: S=1, R=0 followed by S=0, R=0
CASE 2: S=0, R=1 followed by S=0, R=0
CASE 3: S=1, R=1

NOR LATCH TRUTH TABLE:-

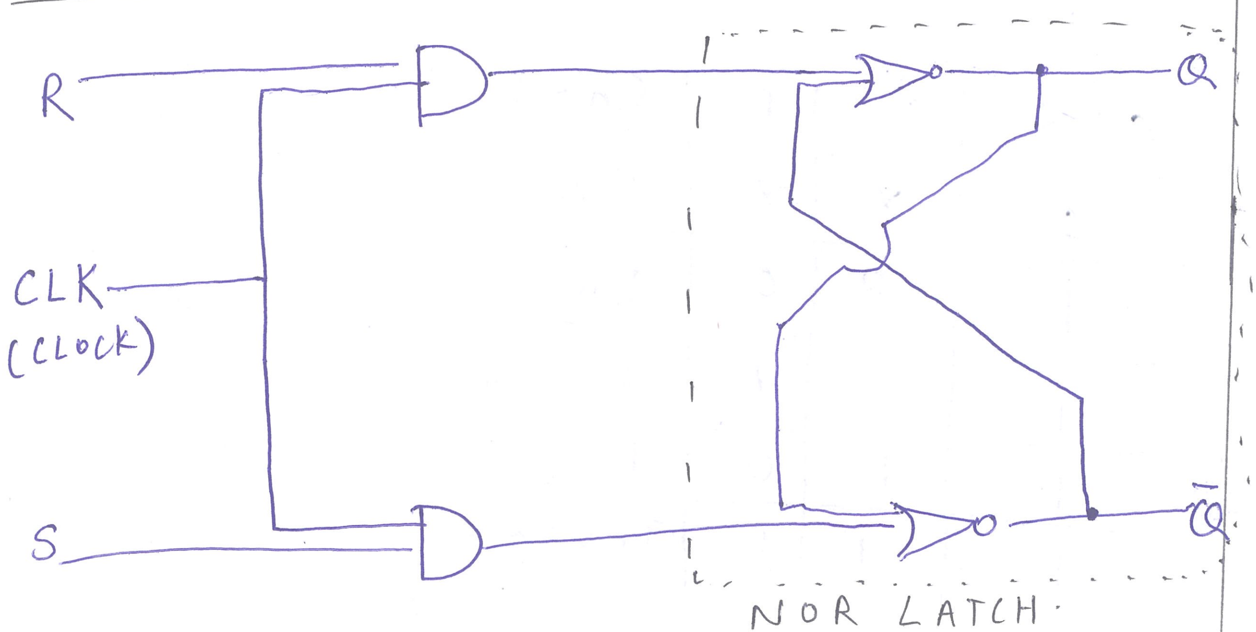
~~S~~A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

for understanding:-

S	R	Q	$\bar{Q}$
0	0	Memory	
0	1	0	1
1	0	1	0
1	1	Invalid	



CLOCKED S-R FLIP-FLOP:-



X - dont care condition.

state varies based on Clock's pulse. output of Q = 0 or 1.

Clock	S	R	Q	$\bar{Q}$
0	X	X	Memory	
1	0	0	Memory	
1	0	1	0	1
1	1	0	1	0
1	1	1	Not used.	

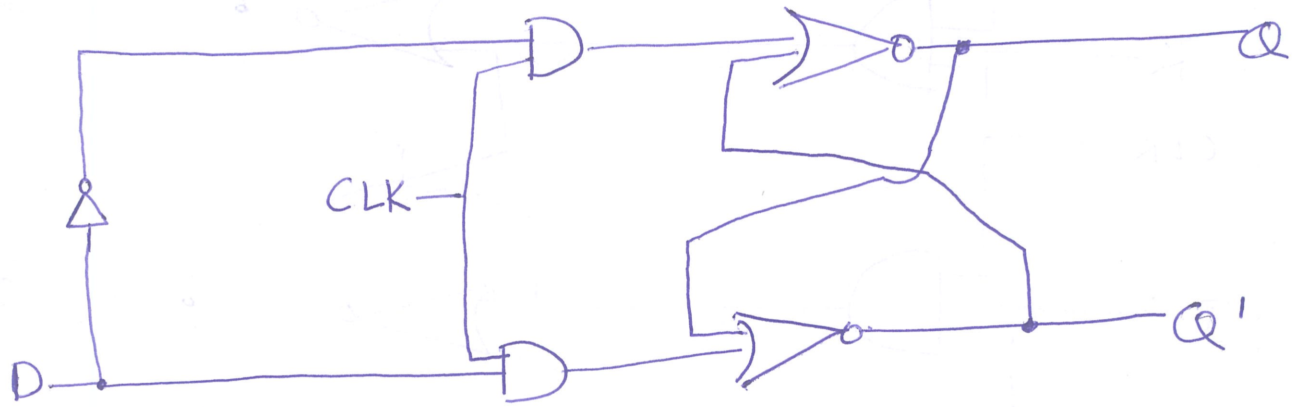
→ Every digital circuit has delay

→ For K-maps, X is taken as 0/1.

The truth Q and $\bar{Q}$ is replaced by Q_{n+1}.

CLK	S	R	Q_{n+1}
0	X	X	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	Invalid

D- Flip Flop



when the clock signal is low, the flip-flop holds current state and ignores the D-input.

→ when the clock signal is high flip-flop samples and stores D input

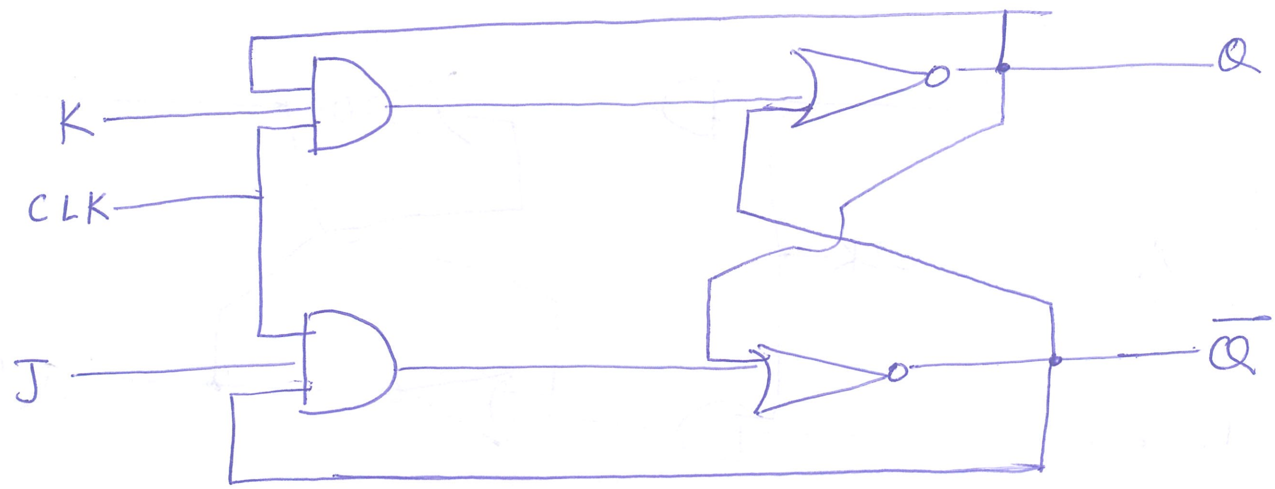
Characteristic equation:-

$$Q(n+1) = D(n)$$

Excitation Table:-

Q(n)	D(n)	Q(n+1)
0	0	0
0	1	1
1	0	0
1	1	1

J-K Flip-Flop:-



The J-K flipflop, unlike S-R flip-flops overcome the problems:-

(i) (S = 0, R = 0) is always avoided.

(ii) if S or R change state while (enable)EN is high, the correct lacthing may not occur.

Characterestic equation:-

$$Q(t+1) = JK' + JQ'(t)$$

J \ KQ_n	$\bar{K}\bar{Q}_n$	$\bar{K}Q_n$	KQ_n	$K\bar{Q}_n$
$\bar{J}$	0	1	0	0
J	1	1	0	1

J \ KQ_n	$\bar{K}\bar{Q}_n$	$\bar{K}Q_n$	KQ_n	$K\bar{Q}_n$
$\bar{J}$	0	1	0	0
J	1	1	0	1

$= \bar{K}Q_n + J\bar{Q}_n$

transforming it, we got

$= JK' + J\dot{Q}'(t)$.

Excitation-table :-

Q(t)	Q(t+1)	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

"x" refers to dont care condition.

Truth Tables

S-R flip-flop

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	–

JK flip-flop

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Q_n

D flip-flop

D	Q_{n+1}
0	0
1	1

Registers :-

A set of n-flip flops used to store n-bits of information.

→ Common clock is used for all flip-flops.

→ A register that provides ability to shift its content is called shift-register.

→ To implement a shift register, it's necessary to use edge triggered or master-slave flip-flops.

There are 4-types of registers:-

(i) PIPO (Parallel In Parallel Out)
(ii) PISO (Parallel In Serial Out)
(iii) SISO (Serial In Serial Out)
(iv) SIPO (Serial In Parallel Out).

Counter:-

A register whose value is easily incremented by 1 modulo the capacity of the register.

- After the Maximum value is acheived, the next increment sets counter value to 0.
- Ex: CPU counter is PC.

Two types:-

Asynchronous/ Ripple Counter	Synchronous Counter.
Relatively slow as the output of flip-flop triggers a change in status of next flip-flop.	- All flip-flops change state at same time. - Because it is faster, it is the kind used in CPUs.

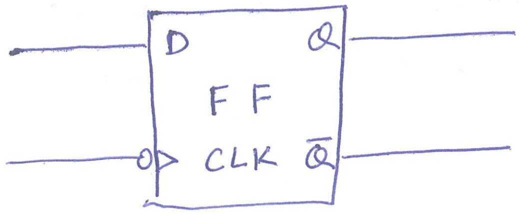
Triggering methods:-

Triggering methods in digital circuits define how flip-flops change their state.

→ One of good triggering methods discussed in our book is negative edge triggering.

P.T.O.

Negative Edge Triggering



The flip-flops changes state only during a clock transistion, on falling (negative edge triggering).

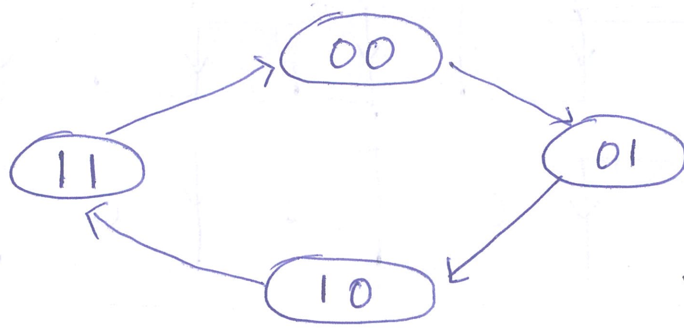
Designing a 2-bit Counter:-

No and type of Flip-Flops:-

Type of flip-flops: J-K
No of flip-flops: 2

Excitation of flip-flop:-

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

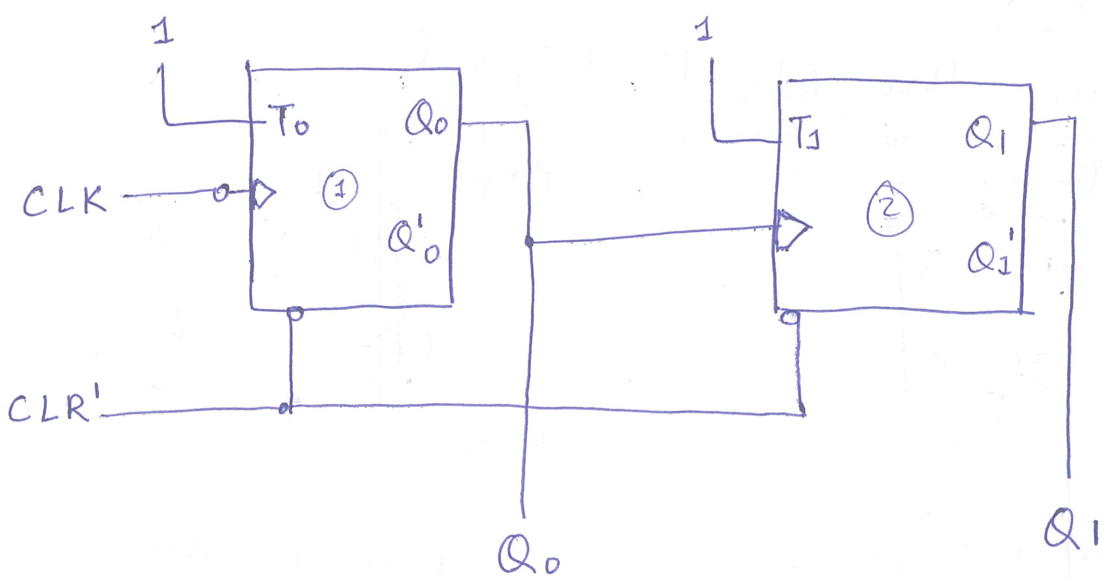


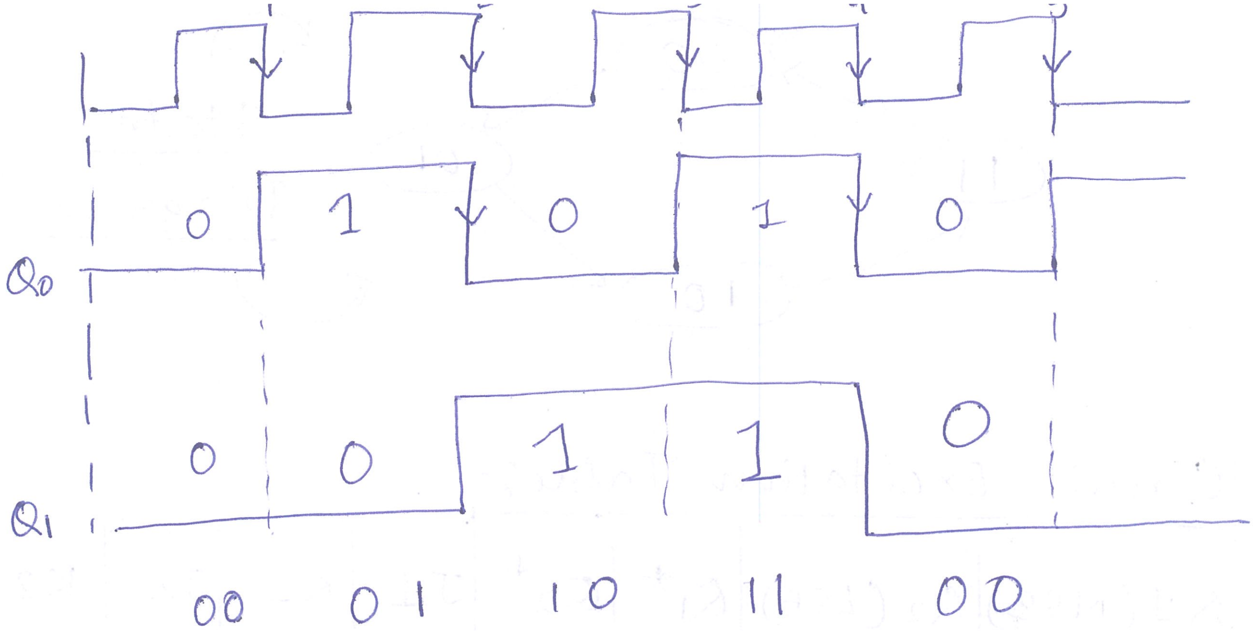
State Diagram

Circuit Excitation Table:-

Q1(MSB)	Q_2(LSB)	Q_1^*	Q_2^*	J1	K1	J2	K2
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	1	1	X
1	1	0	0	X	0	X	1

Logical Diagram:- (2 bit Counter)



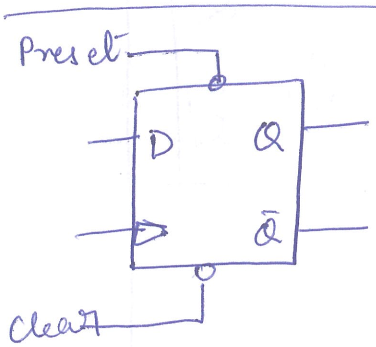


Shift Registers:-

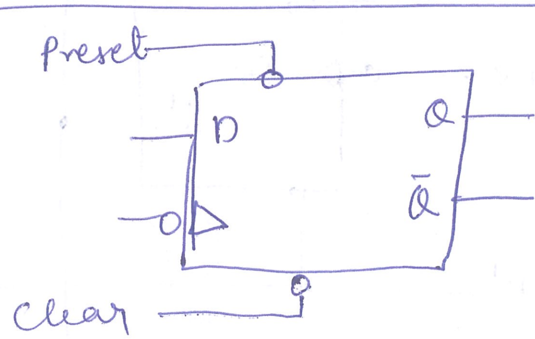
Clears: Asynchronous, Synchronous

Asynchronous Clear:- flip flops are cleared without regard to clock signal.

Synchronous Clear:- Flip-flops are clear with the clock signal.



Positive edge triggered D-flip-flop with Clear and Preset



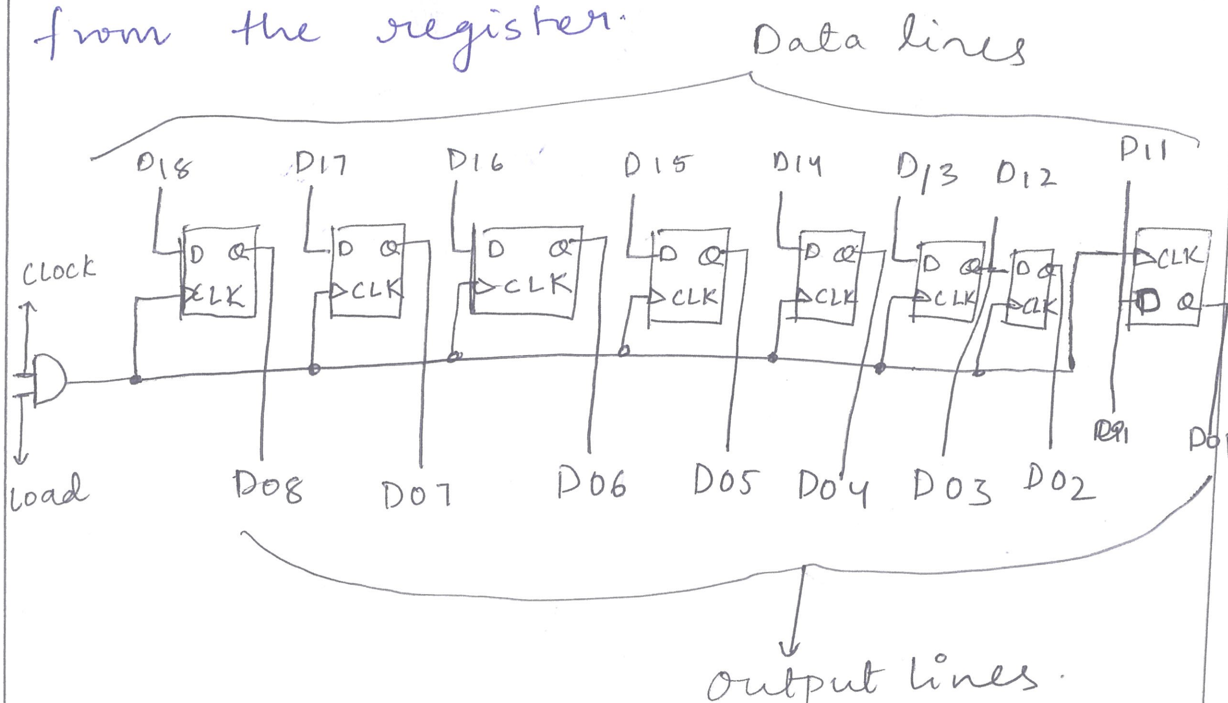
Master slave D-flip-flop with clear and preset.

8-bit Parallel Register:-

It uses PIPO concept.

→ Parallel-In:- This means that all 8-bits can be loaded into register simultaneously, allowing it for fast data transfer.

→ Parallel-out:- This means all 8-bits can also read out of register enabling rapid retrieval of data from the register.



(8-bit parallel register).